A Spectral Diffusion Prior for Unsupervised Hyperspectral Image Super-Resolution

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***Abstract*— Hyperspectral image (HSI) super-resolution with an auxiliary multispectral image (MSI) belongs to the class of inverse problems, where prior knowledge is essential for obtaining the target. Various hand-crafted or deep priors have been developed to enforce the desired solutions. Nevertheless, the spectral distribution knowledge is ignored and still not exploited as a prior. To this end, we design a spectral diffusion model (SDM) to capture the spectral distribution of HSIs and thereby exploit it as a prior for the problem of unsupervised HSI super-resolution. Specifically, we first investigate the spectrum generation problem and extend the diffusion model to fit the 1-D spectral data. Then, we transfer the spectral distribution knowledge of the trained SDM by means of keeping its transition information and thereby induce a regularization term in the framework of maximum a posteriori. At last, we integrate the iterative solving and diffusion generation processes together and employ the Adam to solve the final optimization problem by following the reverse spectral generative sequence. Experimental results conducted on both synthetic and real datasets demonstrate the effectiveness of the proposed approach. The code of the proposed approach is available on https://github.com/liuofficial/SDP.**

***Index Terms*— Deep prior, diffusion model, hyperspectral image (HSI), regularization, super-resolution.**

I. Introduction

HYPERSPECTRAL images (HSIs) can be treated as a kind of 3-D images containing 1-D spectral and 2-D spatial information. Different from the general multichannel images, such as RGB images, HSIs are imaged at hundreds of contiguous and narrow spectral bands, and the spectral range spans the visible to infrared spectrum. Such a high spectral resolution of HSIs makes it possible to accurately identify substances, thus facilitating their practical applications in many fields [1], such as agriculture, mineralogy, and the military. Conversely, the spatial resolution of HSIs is usually relatively low due to the limitation of imaging sensors [2], [3], [4], [5], [6], which hinders their further development. There are two categories of super-resolution methods developed to obtain high-spatial-resolution HSIs (HR-HSIs). One is the fusion-based HSI super-resolution (FusHSISR) that instead records a low-spatial-resolution HSI (LR-HSI) and a conventional high-spatial-resolution image to generate the target HR-HSI. The other is the single HSI super-resolution that performs the task without the auxiliary images. Compared with HSIs, conventional images such as panchromatic and multispectral images (MSIs) can achieve higher spatial resolution with the loss of much spectral information. Benefiting from the complementary information between the two observed images, the FusHSISR has become a satisfactory solution.

The FusHSISR mainly refers to the fusion of a LR-HSI and a high-spatial-resolution MSI (HR-MSI) [3], [4], [5], and the fusion of a LR-HSI and a high-spatial-resolution panchromatic image [2] can be treated as its a special case. This fusion task is similar to the pansharpening that generates an HR-MSI by fusing a low-spatial-resolution MSI and a high-spatial-resolution panchromatic image [7], [8], but it is relatively more complicated due to the need to maintain rich spectral information. Typically, such fusion-based approaches can be roughly classified into four categories: component substitution [9], multiresolution analysis [10], model-based approaches [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], and deep learning-based approaches [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45]. For the FusHSISR, the model-based and deep learning-based approaches have always been the focus of research.

The model-based approaches solve the FusHSISR in terms of an inverse problem by modeling the spectral and spatial degradation processes. They focus on designing effective fidelity terms to characterize the degradation process and exploiting efficient prior information to constrain the range of target solutions. Various fidelity and regularization/constraint terms have been proposed when designing the model-based approaches, and have shown good performance in the FusHSISR [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23]. In the aspect of fidelity terms, many approaches consider using decomposition to depict the intrinsic structure of the target image including unmixing [11], [12], subspace representation [13], [14], dictionary learning [15], [16], and tensor decomposition [17], [18], [19], [20]. In the aspect of prior terms, most approaches are devoted to exploiting spatial, spectral, or spatial–spectral priors to enforce the desired solution. Examples are, that a vector total variation

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regularizer is utilized to measure the piecewise smoothness of an image in the spatial dimension [13]; or that an intracluster self-expressiveness model is exploited to capture the clustering manifold structure of the target image in the spatial domain [21]; or that a structured sparse low-rank representation method is proposed to make use of the high-level spatial and spectral priors [22]; or that a subspace-based low tensor multirank regularization is exploited to depict the spectral correlations and nonlocal similarities of the target image [23].

Deep learning has fueled advancements in remote sensing, particularly with the advent of state-of-the-art large models, portending further enhancement in the intelligent processing of hyperspectral imagery [46], [47]. In the FusHSISR, the deep learning-based approaches mainly consider building a nonlinear mapping between the observed images and the target image. They typically require large amounts of labeled training data and focus on designing complete networks to fit the mapping. There are plenty of excellent networks that have been proposed for the FusHSISR [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45]. Some approaches are devoted to designing fine structures to improve the fusion capability, such as multiscale branches [24], [25], pyramid structure [26], attention mechanism [27], cross-mode information [28], and dense connection [29]. Some approaches improve the ability of the networks to exploit detailed information by injecting high-frequency information [30], [31]. Some approaches consider the dual-stage design strategy and divide the fusion process into the coarse and fine stages [32], [33]. Some approaches introduce the adversarial strategy into the design of networks and thereby improve their robustness [34], [35]. Inspired by model-based approaches, some approaches design networks by unfolding the iterative equations of the models to improve the interpretability of the networks [36], [37]. Considering the drawbacks of convolutional neural networks, some approaches introduce the Transformer architecture into the networks to capture the global relationship and thus improve the fusion quality [38], [39], [40], [41]. Recently, as a new family of deep generative models that are to learn to model the given data distribution, diffusion model [48] has achieved excellent results in many domains such as image and video due to its high generation quality [49]. Motivated by this, some approaches consider modeling the image distribution and adopt conditional diffusion models for the FusHSISR task, taking two observation images as conditional inputs to generate the target image [42], [43], [44], [45].

Nevertheless, both the aforementioned model-based and deep learning-based approaches have their limitations. The model-based approaches with hand-crafted priors rely heavily on manual experience, while the supervised deep learning-based approaches cannot meet the requirement of real scenes well due to the lack of truth reference images. To address this issue, some model-based approaches employ deep networks as priors to improve performance. For instance, Zhang et al. [50] introduced the deep image prior into the fusion model and employed a randomly-initialized neural network to capture the intrinsic information of HSIs. Dian et al. [51] used a pretrained denoising network as a proximal operator and solved the fusion model in an iterative manner. Rui et al. [52] utilized a pretrained diffusion model to characterize image distribution prior and proposed a low-rank factorization strategy for efficient sampling. In the deep learning-based approaches, some researchers mainly draw on autoencoders to implement unsupervised learning. For instance, Qu et al. [53] solved the unsupervised fusion problem via two autoencoder networks, and coupled them through a shared decoder. Wang et al. [54] proposed a variational probabilistic autoencoder consisting of three networks, and jointly optimized these three networks in an unsupervised manner. Gao et al. [55] designed an unmixing-inspired unsupervised network, where two coupled autoencoders are utilized to decompose the two observation images, respectively. Notably, there is no obvious distinction between the model-based approaches using deep priors and the unsupervised deep learning-based approaches. Both of these two kinds of approaches are unsupervised, solve the inverse problems, and use degradation models to construct the loss functions. The unsupervised deep learning-based approaches can be treated as the model-based approaches using specific deep priors [56], and the model-based approaches using deep priors can be solved in an end-to-end manner like the unsupervised deep learning-based approaches [50], [57], [58].

In this article, we desire to exploit a novel deep prior for unsupervised FusHSISR. As mentioned above, various hand-crafted or deep priors have been developed for unsupervised FusHSISR. Notably, as one of the most popular generative models at the moment, the diffusion model has achieved impressive performance in many data modalities including image [48], audio [59], graph [60], and video [61], as well as HSI [42], [43], [44], [45]. The gradient-guided strategy makes it possible to use the diffusion model as a prior [62], [63]. In particular, the image distribution knowledge of HSI has been exploited as a prior via diffusion model [52]. However, there is no prior knowledge of capturing spectral distribution. The motivation for adopting the diffusion model to exploit spectral distribution as a prior is threefold: 1) due to the lack of effective techniques, the importance of spectral distribution knowledge is often ignored; 2) like image and video, a spectrum is a meaningful signal that reflects the absorption properties of different substances; 3) compared with the 3-D cube modality of HSIs or the 2-D image modality, the spectral band number is relatively small, reducing the demand for massive training samples and computational resources.

Motivated by the above issues, we consider the usage of the diffusion model to generate high-quality spectra, and thereby exploit a spectral diffusion prior (SDP) for unsupervised FusHSISR. Specifically, we first propose a spectral diffusion model (SDM) by extending the diffusion model to solve the spectrum generation problem and replace the commonly used UNet denoising network with the more suitable multilayer perceptron (MLP) network to fit the 1-D spectral data. Then, we assume the spectra of target HR-HSI follow the learned spectral data distribution, and employ the trained SDM as a prior by transferring its learned spectral distribution knowledge into the fusion process. The knowledge transfer is achieved by keeping the inverse transition states of SDM, thus inducing a

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regularization term in the framework of maximum a posteriori. At last, in view of the iterative generation process included in SDM, we consider integrating the diffusion iterative generation with the optimization iterative solution and dividing the final optimization problem into a series of subproblems according to the transition process. By treating the target HR-HSI as trainable parameters, we adopt the Adam to solve these subproblems by following the reverse spectral generative sequence in a parameter update manner.

Compared with the existing FusHSISR approaches, some of the innovative characteristics of the proposed SDP are highlighted as follows.

1) For the first time, we capture the spectral distribution knowledge and accordingly exploit a spectral-level prior to guiding the fusion process to generate more accurate spectra. This approach will open a wide field for future developments of spectral distribution knowledge in hyperspectral scenarios.
2) We propose the SDM to model the spectral data distribution of HSIs. It is a pixel-wise model, thus greatly reducing the requirement of training data and computational resources.
3) We exploit the SDM as a prior for FusHSISR, and integrate the spectral diffusion iterative generation into the optimization iterative solution.

The remainder of this article is organized as follows. In Section II, we first introduce the SDM and then present the proposed SDP as well as its optimization algorithm. Section III demonstrates the effectiveness of the proposed approach by conducting several experiments. Section IV concludes this work with some remarks.

II. Proposed Approach

The goal of FusHSISR is to make full use of the complementary information of LR-HSI and HR-MSI to generate the desired HR-HSI. Mathematically, the two observed images can be modeled as spatially degraded and spectrally degraded versions of the target image

$$\mathbf{Y} = \mathbf{XBD} + \mathbf{E}_y \tag{1}$$

$$\mathbf{Z} = \mathbf{RX} + \mathbf{E}_z \tag{2}$$

where $\mathbf{X} \in \mathbb{R}^{N_B \times N_H N_W}$ is the target HR-HSI, with N_B, N_H, and N_W as its spectral band, height, and width, respectively. $\mathbf{Y} \in \mathbb{R}^{N_B \times N_h N_w}$ and $\mathbf{Z} \in \mathbb{R}^{N_b \times N_H N_W}$ are the observed LR-HSI and HR-MSI with N_b ($N_b < N_B$), N_h ($N_h < N_H$), and N_w ($N_w < N_W$) being the multispectral band, low-resolution height, and low-resolution width, respectively. The point spread function (PSF) $\mathbf{B} \in \mathbb{R}^{N_H N_W \times N_H N_W}$ represents the spatial blurring operator, $\mathbf{D} \in \mathbb{R}^{N_H N_W \times N_h N_w}$ represents the spatial downsampling operator, $\mathbf{R} \in \mathbb{R}^{N_b \times N_B}$ represents the spectral response function (SRF) of the multispectral sensor, and $\mathbf{E}_y$ and $\mathbf{E}_z$ are the noises or residuals contained in LR-HSI and HR-MSI.

In the framework of maximum a posteriori, the problem of FusHSISR can be written as

$$\max_{\mathbf{X}} p(\mathbf{X}|\mathbf{Y}, \mathbf{Z}). \tag{3}$$

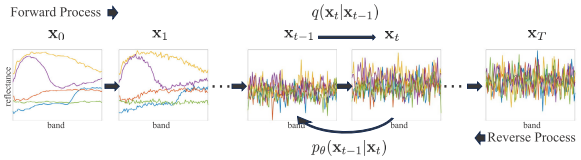


Fig. 1. Illustration of the forward spectral diffusion and reverse spectral generation processes of SDM. From left to right represents the forward process, and from right to left is the reverse process. In each $\mathbf{x}_t$, five spectra of indistinct colors are used as examples.

Using the Bayes formula, the posterior $p(\mathbf{X}|\mathbf{Y}, \mathbf{Z})$ can be rewritten as

$$p(\mathbf{X}|\mathbf{Y}, \mathbf{Z}) = \frac{p(\mathbf{Y}, \mathbf{Z}|\mathbf{X})p(\mathbf{X})}{p(\mathbf{Y}, \mathbf{Z})}. \tag{4}$$

Then, the problem (3) is equivalent as

$$\min_{\mathbf{X}} -\log p(\mathbf{Y}, \mathbf{Z}|\mathbf{X}) - \log p(\mathbf{X}) \tag{5}$$

where the first term is the log-likelihood that depicts the degradation processes of FusHSISR, the second term is the prior (termed as regularization term) that contributes to recovering the target $\mathbf{X}$, and the irrelevant term $p(\mathbf{Y}, \mathbf{Z})$ is dropped. In the optimization, one can write $-\log p(\mathbf{Y}, \mathbf{Z}|\mathbf{X})$ as the following formulation according to (1) and (2)

$$\mathcal{L}_X(\mathbf{X}) = \lambda \|\mathbf{Y} - \mathbf{XBD}\|_F^2 + \|\mathbf{Z} - \mathbf{RX}\|_F^2 \tag{6}$$

where $\lambda > 0$ is a balance parameter, and $\|\cdot\|_F$ represents the Frobenius norm. Then, the objective loss function of FusHSISR can be written as

$$\min_{\mathbf{X}} \mathcal{L}_X(\mathbf{X}) - \log p(\mathbf{X}). \tag{7}$$

In Sections II-A–II-C, we first introduce the SDM to model the spectral data distribution of HSIs, and investigate the spectrum generation quality via Frechet inception distance (FID) [64]. Then, we exploit the SDM as a prior and present the final unified optimization model. At last, we introduce the strategy for solving the model containing the diffusion prior, as well as the algorithmic details.

A. Spectral Diffusion Model

Diffusion model [48] is the new family of deep generative models that can learn the complex distributions over high-dimensional data, and thereby randomly generate samples. As mentioned in Section I, the diffusion model has achieved impressive performance in capturing the data distributions of various modalities. Notably, spectral data contained in HSIs are meaningful signals, and it is feasible to employ a diffusion model to capture their data distribution. However, unlike common data such as images and videos, spectra are a new data modality, and it is necessary to adapt the diffusion model to fit the 1-D spectral data.

The proposed SDM is designed by following the popular denoising diffusion probabilistic model (DDPM) [48]. It consists of two iterative processes, the forward spectral diffusion process and the reverse spectral generation process. Fig. 1 illustrates the two processes of SDM. In the forward process, it gradually adds Gaussian noise to the spectra until they

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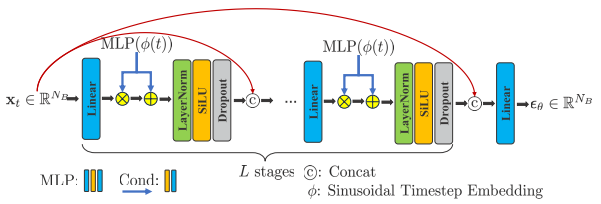


Fig. 2. Architecture overview of the MLP-based denoising network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$.

become random noise. In the reverse process, it starts from Gaussian noise and learns to denoise the spectra by modeling the spectral data distribution.

Suppose we have a data $\tilde{\mathbf{X}}$ that follows the same spectral data distribution as the target HR-HSI $\mathbf{X}$, and for any spectrum $\mathbf{x}_0 \in \mathbb{R}^{N_B}$ within $\tilde{\mathbf{X}}$ it follows the given data distribution $\mathbf{x}_0 \sim q(\mathbf{x}_0)$. As shown in the top of Fig. 1, the forward spectral diffusion process $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ is defined as a Markov chain, which produces latent variables $\{\mathbf{x}_t \in \mathbb{R}^{N_B}\}_{t=1}^T$ by gradually adding Gaussian noise at every timestep t

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}) \tag{8}$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right) \tag{9}$$

where $\{\beta_t\}_{t=1}^T$ is a variance schedule with $0 < \beta_1 < \beta_2 < \cdots < \beta_T < 1$, and $\mathbf{I}$ is the identity matrix. Given a sufficiently large T and well scheduled $\{\beta_t\}_{t=1}^T$, the forward process produces a nearly isotropic Gaussian distribution $\mathbf{x}_T \in \mathbb{R}^{N_B}$. Under the reparameterization trick, one can sample from $q(\mathbf{x}_t|\mathbf{x}_0)$ in a closed form

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \in \mathbb{R}^{N_B} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{10}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{k=1}^{t} \alpha_k$. In the reverse spectral generation process, the estimation of $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is intractable, and a denoising neural network $\boldsymbol{\epsilon}_\theta$ is used to estimate it, where θ represents the network parameters. As shown in the bottom of Fig. 1, the reverse spectral generation process is also defined as a Markov chain that consists of a series of parameterized Gaussian transitions $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) \tag{11}$$

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma_t^2\mathbf{I}\right) \tag{12}$$

where $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$, μ_θ is a learned mean, and σ_t is a fixed variance. For $\mu_\theta(\mathbf{x}_t, t)$, it can be rewritten as

$$\mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) \tag{13}$$

where the denoising network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ is used to estimate the noise contained in $\mathbf{x}_t$. Similar to DDPM, the denoising network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ of SDM can be trained by optimizing the following simple denoising-based objective:

$$\mathcal{L}_\theta(\theta) = \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}, t}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t\right)\right\|^2\right]. \tag{14}$$

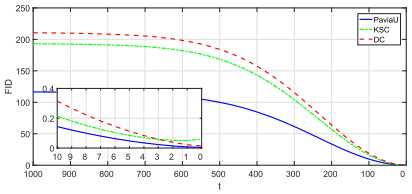


Fig. 3. FIDs as a function of timestep t when applied to the given three datasets.

TABLE I
DETAILED STRUCTURE OF THE MLP-BASED DENOISING NETWORK

Layer		Input dim.	Output dim.
Time embedding ϕ		1	N_ϕ
MLP	Linear	N_ϕ	N_B
	SiLU+Linear	N_B	N_B
Stage 1	Linear	N_B	N_{out}
	Cond: SiLU+Linear	N_B	N_{out}
	LayerNorm+SiLU+Dropout	N_{out}	N_{out}
Stages $2 \cdots L$	Linear	$N_{out} + N_B$	N_{out}
	Cond: SiLU+Linear	N_B	N_{out}
	LayerNorm+SiLU+Dropout	N_{out}	N_{out}
Linear		$N_{out} + N_B$	N_B

In the diffusion model, the standard architecture of denoising network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ is a UNet with time embedding and spatial attentions. One can adjust the UNet to match 1-D spectral data, but it is suboptimal due to the relatively low spectral dimension and the downstream computational requirements. The proposed SDM adopts an MLP-based denoising network, which is shown in Fig. 2. The denoising network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ consists of L stages followed by a *Linear* layer. Each stage outputs a N_{out}-dimensional feature vector and then concatenates it with the input via a skip connection. Similar to other diffusion models, this denoising network is conditioned on the timestep t, which performs sinusoidal timestep embedding ϕ followed by an MLP layer. The time embedding dimension of ϕ is N_ϕ. Table I gives the detailed structure of the MLP-based denoising network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$.

After training the denoising network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ under the loss (14) and plugging the learned noise $\boldsymbol{\epsilon}_\theta$ into μ_θ, one can simply sample $\mathbf{x}_{T-1}, \mathbf{x}_{T-2}, \ldots, \mathbf{x}_0$ by

$$\mathbf{x}_{t-1} = \mu_\theta(\mathbf{x}_t, t) + \sigma_t\boldsymbol{\epsilon}. \tag{15}$$

In image diffusion models, FID is most often used to evaluate the quality of samples. We have found that FID is also suitable for spectral data. In the reverse generative process, the quality of generation spectra gradually becomes better, as the denoising steps are performed. Fig. 3 reports the values of FID at different timestep t during the generative process. Moreover, in the experiments, we show that the spectrum generation quality is positively correlated with the fusion accuracy.

B. Spectral Diffusion Prior

Recent works consider using the gradient-guided strategy to employ the diffusion model as a prior [52], [62], [63]. That is, they generate targets following the reverse iterative generation process, and introduce the gradient information of the reconstruction loss to guide the generation of targets in each iteration. However, this strategy is unsuitable for the proposed SDM due to the relatively complex reconstruction

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loss (6). In this work, we employ the proposed SDM as the prior term $-\log p(\mathbf{X})$ of (7) by transferring its learned spectral distribution knowledge in the framework of maximum a posteriori, and thereby solve it in a model-based way.

We assume that the spectral vectors $\mathbf{x} \in \mathbb{R}^{N_B}$ of the target HR-HSI $\mathbf{X}$ are independent identically distributed, i.e., $-\log p(\mathbf{X}) = -\sum_{\mathbf{x}\in\mathbf{X}} \log p(\mathbf{x})$, and each spectrum follows the spectral data distribution learned by SDM, i.e., $\mathbf{x} = \mathbf{x}_0 \sim q(\mathbf{x}_0)$. In the Markov chain of SDM, the joint distribution of all its states can be written as

$$\log q(\mathbf{x}_{0:T}|\mathbf{x}) = \log\left(q(\mathbf{x}_T|\mathbf{x})\prod_{t=1}^{T} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x})\right) \tag{16}$$

$$= \log q(\mathbf{x}_T|\mathbf{x}) + \sum_{t=1}^{T} \log q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}). \tag{17}$$

The goal of employing SDM as a prior is to transfer the knowledge of $\log q(\mathbf{x}_{0:T}|\mathbf{x})$ into (7). In (17), the first term $\log q(\mathbf{x}_T|\mathbf{x})$ can be omitted as its information is unavailable. The second term $\log q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x})$ is the true logarithmic posteriori distribution that depicts the transition of two neighboring Markov states given $\mathbf{x}$. In the trained SDM, the knowledge is stored in the neural network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ or rather in $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$, and the true posteriori distribution $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ is approximated by $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$. Thus, we introduce a series of constraint terms to embed the spectral distribution knowledge learned by SDM, and the objective loss function (7) of FusHSISR can be written as

$$\begin{aligned} \min_{\mathbf{X}} \quad & \mathcal{L}_X(\mathbf{X}) = \lambda\|\mathbf{Y} - \mathbf{XBD}\|_F^2 + \|\mathbf{Z} - \mathbf{RX}\|_F^2 \\ \text{s.t.} \quad & q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}) = p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) \\ & t \in \{1, \ldots, T\}, \quad \mathbf{x} \in \mathbf{X}. \end{aligned} \tag{18}$$

The optimization problem (18) is a constrained one, and we can rewrite it as the following unconstrained form:

$$\min_{\mathbf{X}} \mathcal{L}_X(\mathbf{X}) + \gamma \sum_{t,\mathbf{x}} \mathcal{L}_{kl}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x})||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) \tag{19}$$

where $\gamma > 0$ is a regularization parameter, and $\mathcal{L}_{kl}$ denotes the Kullback–Leibler (KL) divergence. Under the assumption of $\mathbf{x}$, the second term of (19) can be rewritten as the form of (14) using the reparameterization trick

$$\mathcal{L}_\theta(\mathbf{x}, t) = \mathbb{E}_{\boldsymbol{\epsilon}}\left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x} + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t\right)\|^2\right] \tag{20}$$

where the weight that is a function of the noise schedule is omitted. With the above derivation, the final objective loss function of the proposed SDP can be formulated as

$$\min_{\mathbf{X}} \mathcal{L}_X(\mathbf{X}) + \gamma \sum_{t,\mathbf{x}} \mathcal{L}_\theta(\mathbf{x}, t). \tag{21}$$

C. Optimization

The final objective (21) consists of an ordinary fidelity term and an SDM-based regularization term. The regularization term is originally used to train the network $\boldsymbol{\epsilon}_\theta$ of the proposed SDM, and its network parameters θ are fixed in solving (21). To solve (21), we should optimize the objective with respect to $\mathbf{X}$ through sampling $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ in the summands over t.

Algorithm 1 SDP

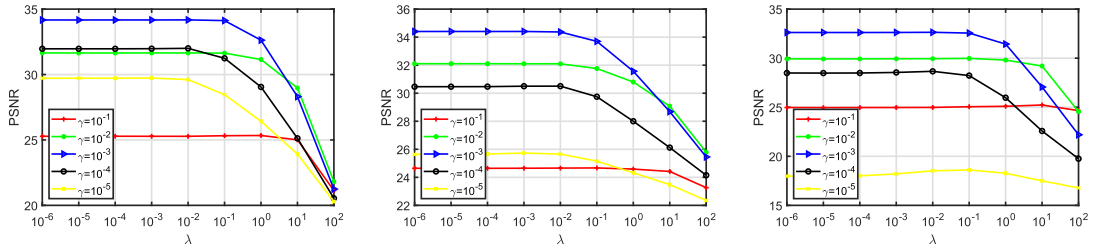
Input: LR-HSI $\mathbf{Y}$, HR-MSI $\mathbf{Z}$, clean hyperspectral pixels $\tilde{\mathbf{X}}$, PSF $\mathbf{B}$, SRF $\mathbf{R}$, spatial downsampling $\mathbf{D}$, regularization parameters λ and γ, learning rate μ.
Train the loss (14) of the MLP-based denoising network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ in SDM by randomly feeding the spectra of $\tilde{\mathbf{X}}$.
Initialize the parameters $\mathbf{X}$ and fix the network parameters θ.
for $t = T : 1$ **do**
  for $k = 1 : K$ **do**
    Sample $\boldsymbol{\epsilon} \in \mathbb{R}^{N_B \times N_H N_W} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
    $\mathbf{X}_t = \sqrt{\bar{\alpha}_t}\mathbf{X} + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$
    Compute $\boldsymbol{\epsilon}_\theta \in \mathbb{R}^{N_B \times N_H N_W}$ by feeding $\mathbf{X}_t$ and t
    $\mathbf{X} \leftarrow \mathbf{X} - \mu\nabla_{\mathbf{X}}[\mathcal{L}_X(\mathbf{X}) + \gamma\sum_{\mathbf{x}} \mathcal{L}_\theta(\mathbf{x}, t)]$
  end for
end for
Output: HR-HSI $\mathbf{X}$.

The SDM gradually generates spectra using (15) during the sampling process. Intuitively, a large t should coarsely generate $\mathbf{x}_0 = (\mathbf{x}_t - (1-\bar{\alpha}_t)^{1/2}\boldsymbol{\epsilon}_\theta)/(\bar{\alpha}_t)^{1/2}$, and it is impractical to optimize (21) simultaneously for all t. Similar to the sampling process, we sequentially solve the tth subproblem of (21) from T to 1, so as to match the iterative solutions in optimization. Specifically, we treat $\mathbf{X}$ as trainable parameters, adopt the Adam as the optimizer, and perform the gradient update K times for each t-subproblem. The complete process of the proposed SDP is summarized in Algorithm 1. Notably, when K is set to be 1, this algorithm will degenerate into the existing gradient-guided ways [62], [63]. In this work, K should be larger than 1 as demonstrated in the experiments.

III. Experimental Results and Analysis

This section conducts experiments on both synthetic and real datasets to evaluate the performance of the proposed SDP. All datasets are scaled to the range [0, 1]. For the synthetic datasets, five metrics are used to assess the quality of the fused images, namely, peak signal-noise-ratio (PSNR), spectral angle mapper (SAM), relative dimensionless global error in synthesis (ERGAS), universal image quality index (UIQI), and structural similarity (SSIM) [2], [3], [4]. These metrics evaluate the quality of the fused images from different perspectives. PSNR measures the similarity between the target and reference images at the pixel level. SAM and SSIM assess the quality of fused images from spectral and spatial perspectives, respectively. ERGAS considers the mean square error and root mean square error as well as the brightness information to provide a comprehensive assessment of the fusion performance. UIQI provides the evaluation of the fused images in three aspects: correlation loss, brightness distortion, and contrast distortion. For the real dataset, three metrics without reference are adopted to assess the quality of the fusion results, namely, spectral distortion index D_λ, spatial distortion index D_s, and quality with no reference (QNR) [8], [65].

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(a) (b) (c)

Fig. 4. PSNR with respect to parameters λ and γ. (a) PaviaU dataset. (b) KSC dataset. (c) DC dataset.

A. Synthetic Datasets and Implementation Details

Given three real-life HSIs, University of Paiva (PaviaU), Kennedy Space Center (KSC), and Washington DC Mall (DC), we take the source image as the reference image and generate the corresponding two observation images LR-HSI and HR-MSI, according to the Wald's protocol [66]. The three real-life HSIs are listed as follows.

1) The PaviaU dataset is captured by the Reflective Optics System Imaging Spectrometer (ROSIS). This image has 115 spectral bands covering the spectral range 0.43–0.86 μm, and has 610 $\times$ 340 pixels with a spatial resolution of 1.3 m per pixel. After the removal of noisy bands, it has 103 spectral bands remained. We crop the up-left 512 $\times$ 256-pixel part for experiments.
2) The KSC dataset is captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS). This image has 224 spectral bands covering the spectral range 0.4–2.5 μm, and has 512 $\times$ 614 pixels with a spatial resolution of 18 m per pixel. The number of its spectral bands is reduced to 176 by removing water absorption bands. We crop the up-left 512 $\times$ 256-pixel part for experiments.
3) The DC dataset is captured by the Hyperspectral digital imagery collection experiment (HYDICE) image. This image has 210 spectral bands covering the spectral range 0.4–2.4 μm, and has 1208 $\times$ 307 pixels with a spatial resolution of about 2.8 m per pixel. There are 191 bands left by removing bands in the specific region, where the atmosphere is opaque. We crop the up-left 512 $\times$ 256-pixel part for experiments.

We divide the cropped image into two 256 $\times$ 256-pixel parts. The top one is used as the clean hyperspectral pixels $\tilde{\mathbf{X}}$, and the bottom one is used as the reference image $\mathbf{X}$. The synthetic processes of the observation images LR-HSI and HR-MSI can be depicted by (1) and (2), respectively. For the LR-HSI, the PSF $\mathbf{B}$ is achieved by spatially blurring each band of the reference image, where a Gaussian blur of size 15 $\times$ 15 with mean 0 and standard deviation 3.40 is adopted. The operator $\mathbf{D}$ is achieved by spatially downsampling the blurred image with a factor of 8 in two directions. The noise $\mathbf{E}_y$ is achieved by adding 20 dB Gaussian noise to the LR-HSI. For the HR-MSI, the SRF $\mathbf{R} \in \mathbb{R}^{4\times N_B}$ is derived from the spectral response of the IKONOS satellite, according to the spectral response profiles of the RGB and NIR bands. A Gaussian noise with 30 dB is added to the HR-MSI.

TABLE II
IMPLEMENTATION DETAILS OF SDM

Parameter	Value
Stage L	4
Stage output dim. N_{out}	512
Time embedding dim. N_ϕ	64
Dropout rate	0.001
Total diffusion steps T	1000
β scheduler	Linear
β_1, β_T	0.0001, 0.02
Optimizer	Adam
Batch size	512
Epoches	30000
Learning rate	0.01
Learning rate scheduler	$0.001 \times \max(1000 - epoch/10, 1)$

TABLE III
PSNR AS A FUNCTION OF TOTAL NUMBER T OF DIFFUSION STEPS WHEN APPLIED TO THE GIVEN THREE DATASETS

Steps T	500	750	1000	1500	2000	3000
Pavia	28.97	33.74	34.12	33.98	33.94	33.94
KSC	25.56	32.82	33.70	33.80	33.63	33.64
DC	30.53	32.14	32.56	32.24	32.21	32.04

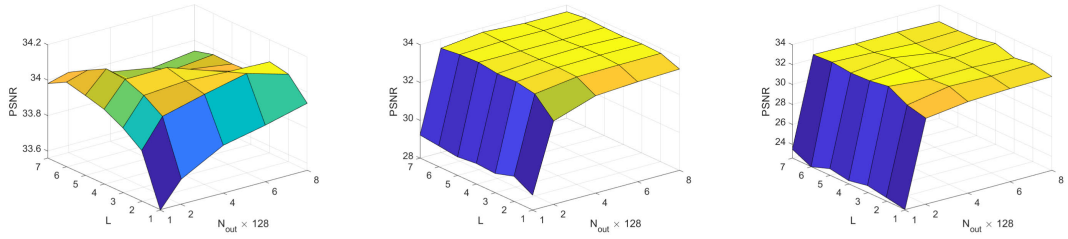
We implement the SDM and SDP in the PyTorch framework. The implementation details of the proposed SDM are provided in Table II. For SDP, the inner loop K is set to be 3, and the learning rate μ is set as 0.001, 0.001, and 0.0025 for the PaviaU, KSC, and DC datasets, respectively.

B. Parametric Analysis and Ablation Study

1) Influence of λ and γ: There are two parameters λ and γ that need to be tuned in the proposed SDP. This set of experiments is to investigate them and show their impact on the fusion results. Fig. 4 illustrates the influence of these two parameters in terms of PSNR. It can be observed that the results of $\gamma = 10^{-3}$ are the best for all datasets. The choice of λ has a wide optimal range in $[10^{-6}, 10^{-1}]$, which indirectly demonstrates the effectiveness of the proposed prior. Thus, λ and γ are eventually set as 10^{-1} and 10^{-3} for all datasets.

2) Ablation Study: This set of experiments is to investigate the architecture of the proposed SDM and SDP. In SDM, the model size of the MLP-based denoising network $\boldsymbol{\epsilon}_\theta$ can be controlled by adjusting the stage L and the output dimension of each stage N_{out}. Fig. 5 shows the PSNR with respect to these two variables. It can be seen that when L and N_{out} are relatively small, the fusion accuracy is low. For the PaviaU dataset, the fusion accuracy first increases and then decreases, as L increases; while for the other two datasets,

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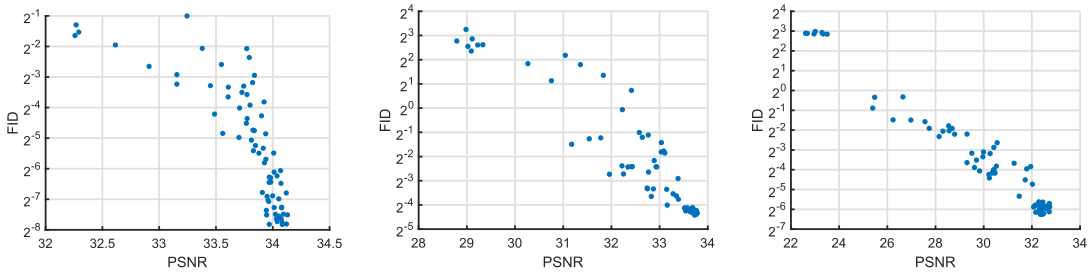
(a) (b) (c)

Fig. 5. PSNR with respect to L and N_{out}. (a) PaviaU dataset. (b) KSC dataset. (c) DC dataset.

TABLE IV

DIFFERENT COMBINATIONS OF DENOISING NETWORKS AND INNER ITERATIONS (THE BEST VALUES ARE MARKED IN BOLD)

Setting						PaviaU			KSC			DC		
Case	MLP	UNet	$K=1$	$K=3$	$K=5$	PSNR	SAM	Time	PSNR	SAM	Time	PSNR	SAM	Time
1						27.88	7.98	**6.3s**	21.97	19.34	**10.9s**	15.35	20.41	**12.1s**
2		√	√			20.33	10.01	296.9s	15.97	12.21	1.3h	27.75	5.42	1.2h
3		√		√		34.12	**3.05**	0.2h	33.73	4.15	4.4h	31.58	2.56	5.2h
4		√			√	33.92	3.11	0.4h	33.66	4.19	6.8h	31.32	**2.56**	6.3h
5	√		√			20.32	10.02	28.3s	16.02	12.25	33.4s	27.47	5.33	35.8s
6	√			√		**34.12**	3.07	120.2s	33.70	4.11	145.1s	**32.56**	2.61	150.4s
7	√				√	34.03	3.10	137.3s	**33.76**	**4.05**	160.1s	32.44	2.59	169.1s



(a) (b) (c)

Fig. 6. Illustration of the relationship between FID and PSNR. (a) PaviaU dataset. (b) KSC dataset. (c) DC dataset.

the fusion accuracy first increases and then tends to saturation. As for N_{out}, one can observe similar results when N_{out} is varied.

The total number T of diffusion steps affects the reverse spectral generation of SDM and the gradient optimization of SDP. Table III presents the PSNRs of the proposed method under different total diffusion steps T. It can be observed that for all datasets, as T increases, the PSNR first rises and then tends to saturate.

In SDM, the commonly used UNet denoising network is replaced with the MLP-based one. In SDP, there exists a number K of inner iterations to be adjusted. Table IV reports the results of different combinations of these two elements. When all elements are not selected (case 1), it represents that no spectral prior is introduced into the proposed SDP, i.e., the problem (21) with $\gamma = 0$. When MLP or UNet is selected, the corresponding architecture is adopted as the denoising network. It can be seen that using MLP or UNet as a denoising network has similar accuracy, but using UNet takes more computation time. Cases 3, 4, 6, and 7 with the introduction of spectral priors outperform case 1 without the introduction of spectral prior. The performance when K is greater than 1 is significantly better than the performance when $K = 1$. For the PaviaU and KSC datasets, the performance when $K = 1$ is even lower than case 1 where no spectral prior is introduced.

C. Experiment Results on Synthetic Datasets

1) Relationship Between FID and Fusion Accuracy: This set of experiments is to show how the spectrum generation quality FID affects the fusion accuracy. By varying stage L, output dimension N_{out}, and training epoch, we train 71 different SDMs and thereby compute the FIDs of their generated spectra. Then, for each model, we perform SDP by adopting it as a prior. Fig. 6 illustrates the FIDs of these SDMs and the PSNRs of their corresponding fusion results. It is clear that in the general trend, the higher the generation quality of the SDM, the better the fusion result with it as a prior.

2) Comparison With the State-of-the-Art: This set of experiments is to evaluate the performance of the proposed SDP by comparing it with eight related methods. These methods can be divided into two categories. The first category consists of four model-based methods using hand-craft priors, i.e., CSU [12], HySure [13], NSSR [15], and HCTR [18]. The second category is the four unsupervised (model-based or deep learning-based) methods, including CNNFUS [51], MIAE [56], EU2ADL [55], and PLRDiff [52]. The free parameters of these comparison methods are tuned on our datasets. For all comparison methods, they are performed in an undefined condition, that is, the PSF $\mathbf{B}$ and SRF $\mathbf{R}$ are unknown. The estimation network proposed in [56] is adopted to estimate these two functions.

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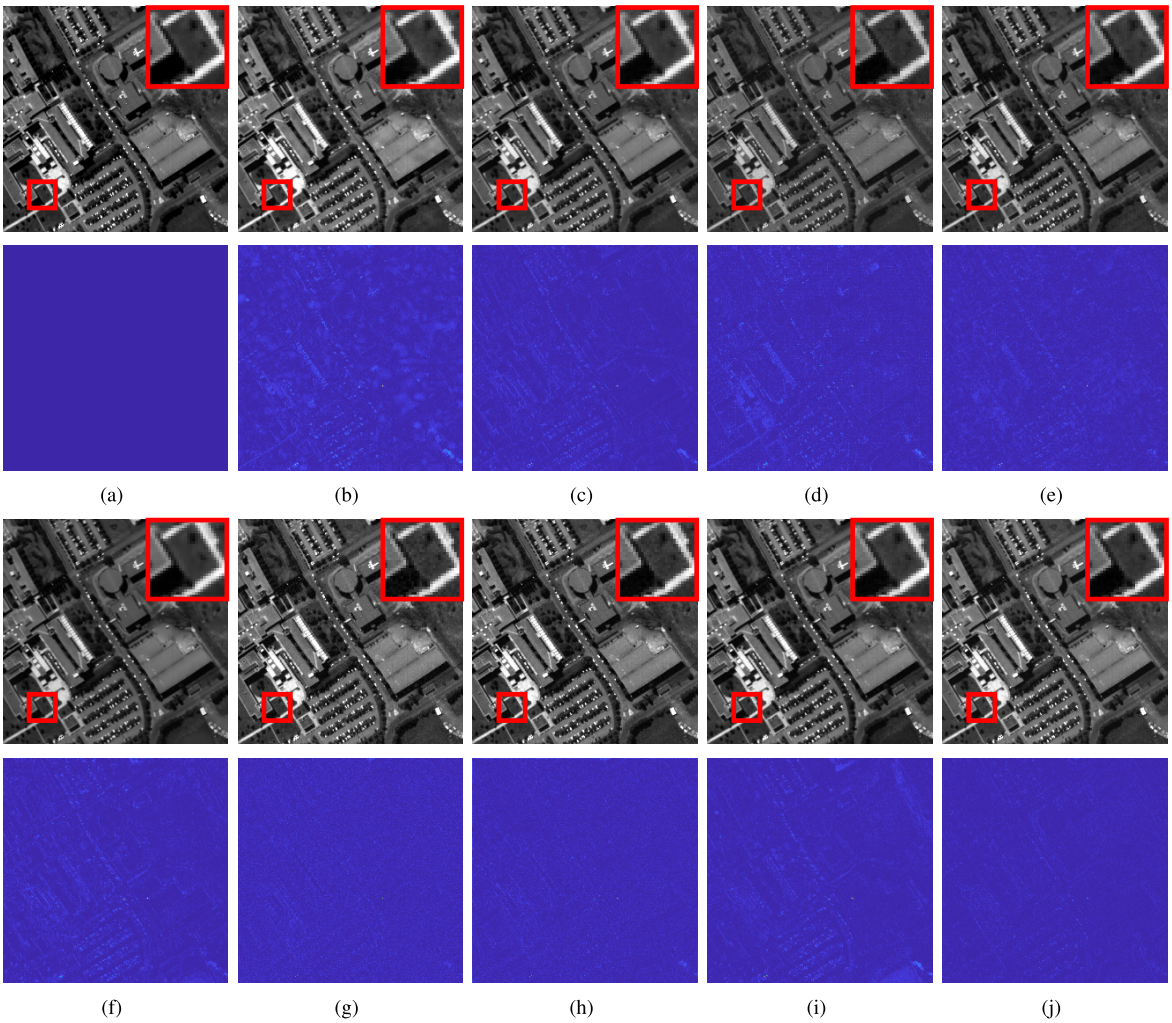


Fig. 7. Fused images and error maps at the 30th band when applied to the PaviaU dataset. (a) Reference image. (b) CSU. (c) HySure. (d) NSSR. (e) HCTR. (f) CNNFUS. (g) MIAE. (h) EU2ADL. (i) PLRDiff. (j) SDP.

TABLE V
COMPARISON OF DIFFERENT METHODS ON THE PAVIAU DATASET (THE BEST VALUES ARE MARKED IN BOLD)

Method	PSNR	SAM	ERGAS	UIQI	SSIM
Best Values	$+\infty$	0	0	1	1
CSU	29.98	4.34	1.217	0.971	0.931
HySure	32.05	3.81	1.029	0.979	0.943
NSSR	29.74	4.57	1.206	0.971	0.918
HCTR	32.05	4.06	1.045	0.979	0.941
CNNFUS	32.37	3.54	0.981	0.980	0.944
MIAE	32.27	3.93	1.001	0.978	0.929
EU2ADL	32.06	3.76	0.993	0.980	0.936
PLRDiff	31.98	4.37	1.148	0.977	0.937
SDP	**34.12**	**3.07**	**0.825**	**0.984**	**0.949**

TABLE VI
COMPARISON OF DIFFERENT METHODS ON THE KSC DATASET (THE BEST VALUES ARE MARKED IN BOLD)

Method	PSNR	SAM	ERGAS	UIQI	SSIM
Best Values	$+\infty$	0	0	1	1
CSU	27.02	6.76	2.614	0.695	0.886
HySure	31.98	4.56	1.929	**0.826**	0.938
NSSR	28.70	7.22	2.527	0.707	0.877
HCTR	31.56	6.39	2.211	0.755	0.914
CNNFUS	30.93	4.38	1.953	0.821	0.942
MIAE	30.27	6.51	2.375	0.726	0.885
EU2ADL	31.10	6.57	2.173	0.731	0.891
PLRDiff	31.77	7.10	2.046	0.723	0.876
SDP	**33.70**	**4.11**	**1.787**	0.805	**0.943**

Table V reports the five quantitative results of the comparison methods on the PaviaU dataset. As seen from this table, the proposed SDP method performs the best among the comparison methods followed by CNNFUS, and its improvements over the second CNNFUS are near 2 dB in PSNR and 0.5° in SAM. Fig. 7 illustrates the fused images and their error images with respect to the reference image, in terms of the 30th band. The reference image is also included for reference. Visually, it can be seen from the fused images that most methods generate high-quality images. From the error images, it can be observed that SDP, MIAE, and EU2ADL are closer to the reference image. Fig. 10(a) presents the PSNR of each spectral band for the comparison methods. In this figure, the proposed SDP method gives the best values in almost all bands. Fig. 11(a) presents the SAM of each pixel for the comparison methods, with the pixels sorted in order of ascending error. It can be seen that SDP consistently outperforms the other methods at the pixel level followed by CNNFUS.

Table VI presents the PSNR, SAM, ERGAS, UIQI, and SSIM results of the comparison methods on the KSC dataset.

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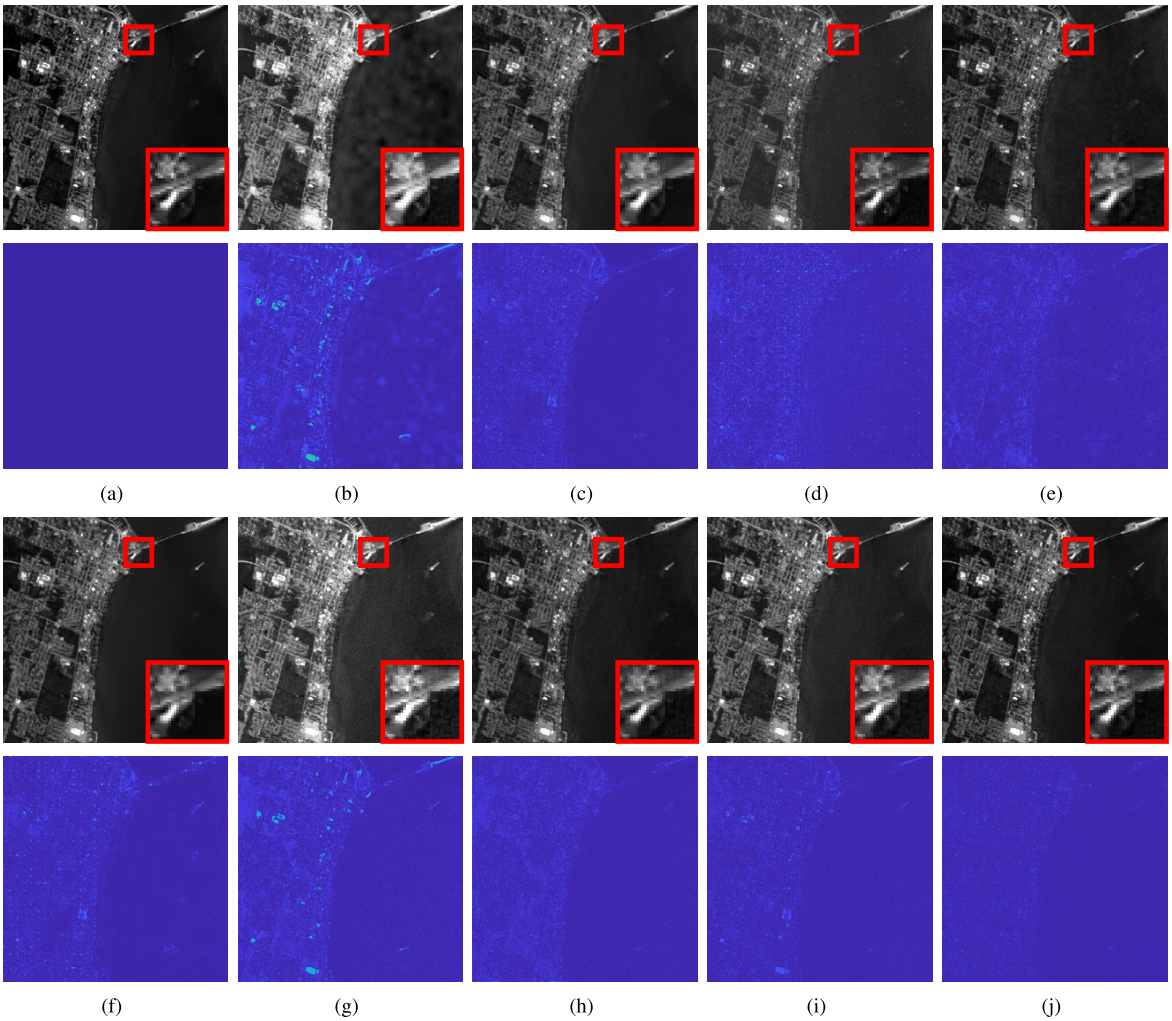


Fig. 8. Fused images and error maps at the 30th band when applied to the KSC dataset. (a) Reference image. (b) CSU. (c) HySure. (d) NSSR. (e) HCTR. (f) CNNFUS. (g) MIAE. (h) EU2ADL. (i) PLRDiff. (j) SDP.

TABLE VII
COMPARISON OF DIFFERENT METHODS ON THE DC DATASET (THE BEST VALUES ARE MARKED IN BOLD)

Method	PSNR	SAM	ERGAS	UIQI	SSIM
Best Values	$+\infty$	0	0	1	1
CSU	27.73	3.76	11.215	0.813	0.938
HySure	27.85	4.65	12.051	0.823	0.919
NSSR	25.62	7.75	11.358	0.817	0.898
HCTR	27.74	5.70	11.615	0.858	0.915
CNNFUS	27.81	4.91	11.483	0.857	0.921
MIAE	29.53	3.59	10.048	0.875	0.951
EU2ADL	28.45	3.68	11.135	0.892	0.944
PLRDiff	27.10	5.31	37.672	0.850	0.899
SDP	**32.56**	**2.61**	**9.524**	**0.959**	**0.973**

It can be seen from this table that the CSU method without a prior performs the worst in PSNR, ERGAS, and UIQI. The proposed SDP method outperforms the other methods in four metrics, and its improvements are over 1.7 dB in PSNR and 0.2° in SAM. The fused images and their error maps at the 30th band are shown in Fig. 8. From this figure, it can be observed that the CSU method generates the worst results, and the results of SDP are better than the others. To compare the aforementioned methods in different respects, Figs. 10(b) and 11(b) give the PSNR and SAM as functions of the spectral band and sorted pixel, respectively. It can be seen that the proposed SDP method can give the best results in most cases.

Table VII summarizes the results of the comparison methods under the five quality metrics for the DC dataset. It can be seen from this table that many methods do not get satisfactory results, and the proposed SDP method achieves the highest results in all metrics. The improvements of SDP over the second method (MIAE) are about 3 dB in PSNR and 1° in SAM. Fig. 9 illustrates the fusion results and error maps of the comparison methods in terms of the 30th band. Through visual inspection, it can be seen that there is obvious spatial distortion in the CSU method, and the results of SDP are closer to the truth. Figs. 10(c) and 11(c) illustrate the PSNR of each spectral band and the SAM of each sorted pixel, respectively. These two figures report that SDP achieves higher results than the others in terms of band-level PSNR and pixel-level SAM.

3) Computational Efficiency: The experiments are carried out using a desktop computer with an Intel Core i9-13900K CPU, a GeForce RTX 4090 GPU, and 64 GB memory. There are four methods that are implemented by the PyTorch framework, i.e., MIAE, EU2ADL, PLRDiff, and

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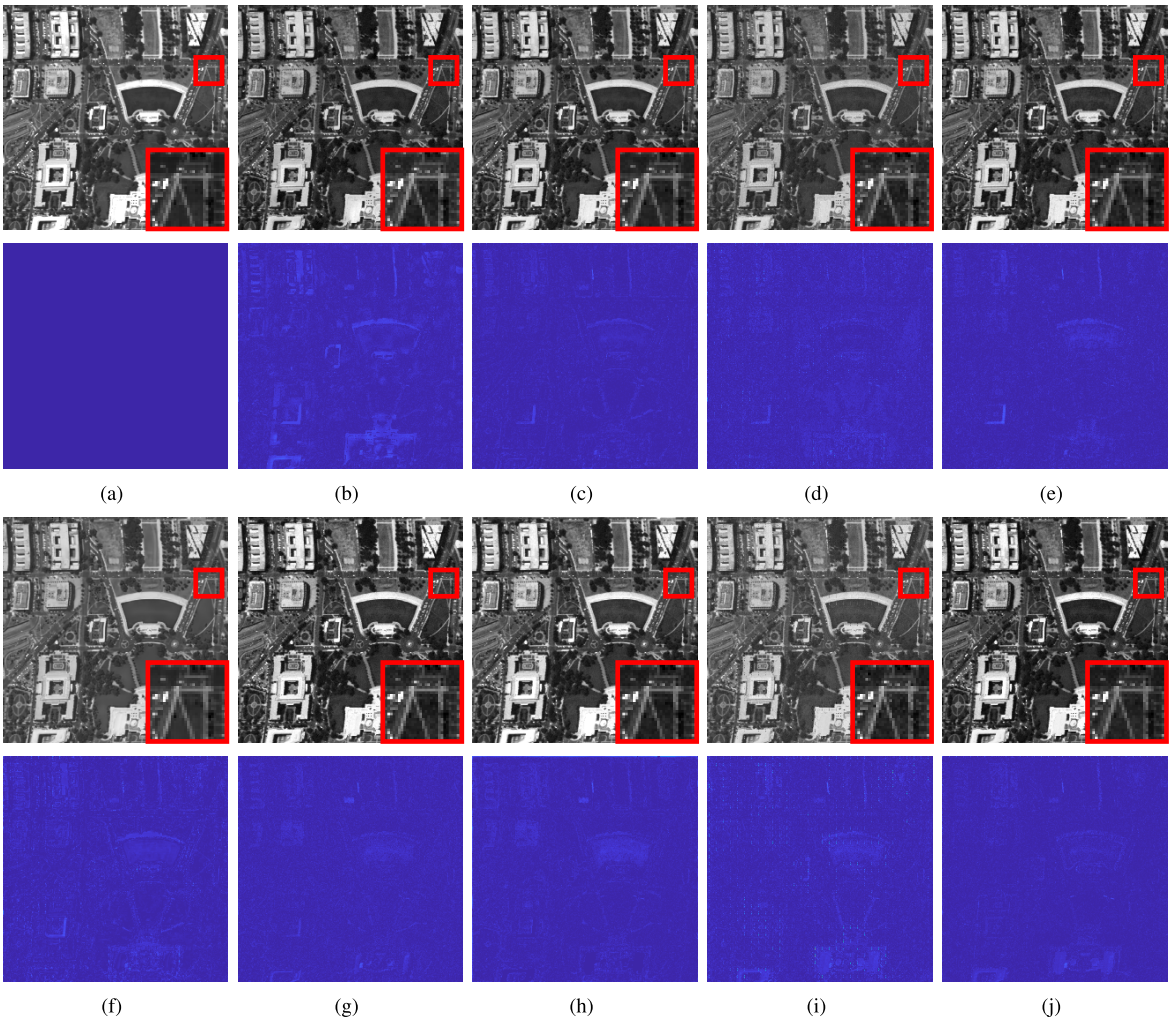


Fig. 9. Fused images and error maps at the 30th band when applied to the DC dataset. (a) Reference image. (b) CSU. (c) HySure. (d) NSSR. (e) HCTR. (f) CNNFUS. (g) MIAE. (h) EU2ADL. (i) PLRDiff. (j) SDP.

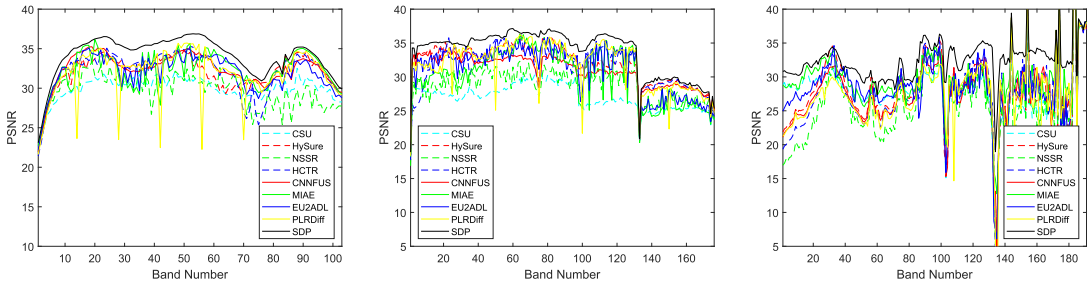


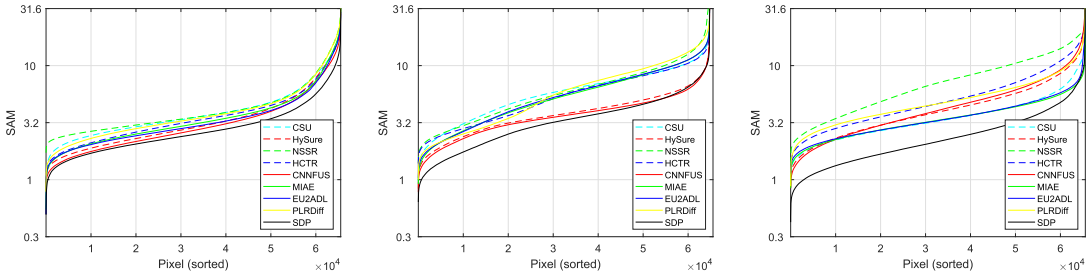
Fig. 10. PSNR as a function of the spectral band. (a) PaviaU dataset. (b) KSC dataset. (c) DC dataset.

SDP. The remaining methods CSU, HySure, NSSR, HCTR, and CNNFUS are performed using MATLAB. Table VIII summarizes the running time of these comparison methods as a reference, as well as the training time of the proposed SDM.

D. Experiment Results on Real Data

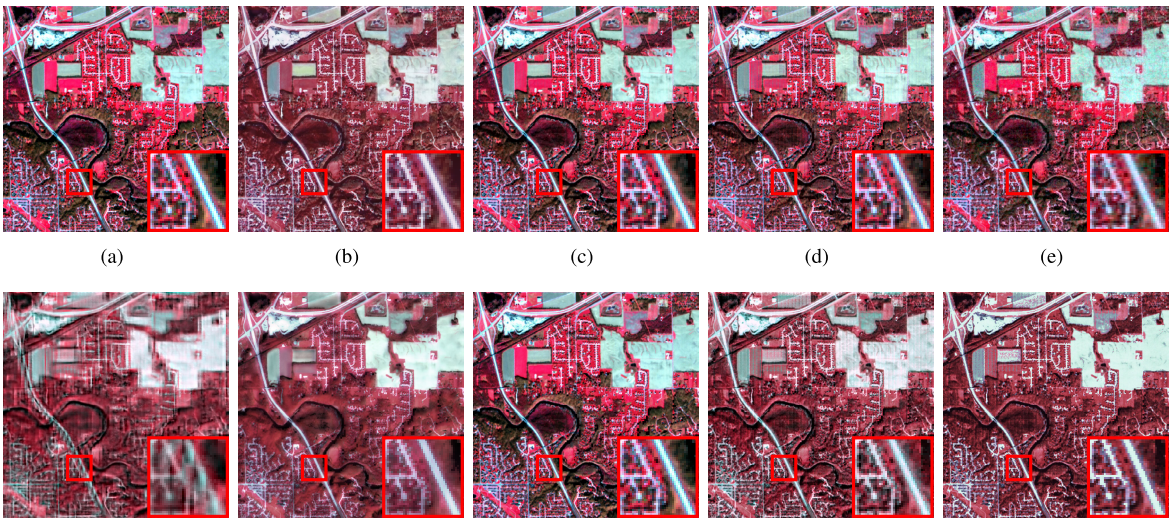
A real spaceborne dataset [67] is used to evaluate SDP in practical application. The LR-HSI is acquired by the Hyperion sensor carried on the Earth Observing-1 satellite, and the HR-MSI is acquired by the Sentinel-2A satellite. These two images are taken over Lafayette, LA, USA, in October and November 2015, respectively. The Hyperion sensor is characterized by 242 spectral bands, with a spectral range of 0.4 to 2.5 μm. The noisy bands and water absorption bands are removed, leaving 83 bands. The spatial resolution of LR-HSI is 30 m. The HR-MSI has 13 bands, where four bands with 10 m spatial resolution are kept. The LR-HSI is spatially downsampled with a factor of 2 in two directions. Then, we crop a 64 $\times$ 64-pixel HSI and a 384 $\times$ 384-pixel MSI as observation images from the overlapped region of the LR-HSI and HR-MSI, respectively. The LR-HSI is used as the clean hyperspectral pixels. Table IX reports the results of three metrics without reference for the comparison methods

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(a) (b) (c)

Fig. 11. SAM (plotted in a $\log_{10}(\cdot)$ scale) as a function of sorted pixel. (a) PaviaU dataset. (b) KSC dataset. (c) DC dataset.



(a) (b) (c) (d) (e)

(f) (g) (h) (i) (j)

Fig. 12. False-color images of fusion results when applied to the real Hyperion-Sentinel dataset. (a) HR-MSI. (b) CSU. (c) HySure. (d) NSSR. (e) HCTR. (f) CNNFUS. (g) MIAE. (h) EU2ADL. (i) PLRDiff. (j) SDP.

TABLE VIII

RUNNING TIME (IN SECONDS) OF THE COMPARISON METHODS AND TRAINING TIME OF SDM

Method	PaviaU	KSC	DC
CSU	38.1	12.6	23.4
HySure	45.1	44.9	45.0
NSSR	26.2	40.5	44.9
HCTR	17.9	27.2	30.6
CNNFUS	1.5	2.0	1.9
MIAE	39.3	57.9	36.2
EU2ADL	301.2	329.0	186.4
PLRDiff	145.5	146.1	146.7
SDM	170.6	171.4	173.7
SDP	120.2	145.1	150.4

TABLE IX

QUALITY METRICS WITHOUT REFERENCE FOR THE REAL HYPERION-SENTINEL DATASET (THE BEST VALUES ARE MARKED IN BOLD)

Method	D_λ	D_s	QNR
Best	0	0	1
CSU	0.0518	0.1136	0.8405
HySure	0.1271	0.0975	0.7878
NSSR	0.1353	0.1472	0.7374
HCTR	0.0598	**0.0422**	0.9005
CNNFUS	0.0567	0.1136	0.8362
MIAE	0.0409	0.0708	0.8912
EU2ADL	0.0720	0.0679	0.8650
PLRDiff	0.0593	0.0699	0.8749
SDP	**0.0228**	0.0704	**0.9084**

mentioned in Section III-C. It can be seen that the proposed SDP gives the highest QNR and D_λ, and HCTR gives the highest D_s. Fig. 12 illustrates the false-color images of the observation images and the fusion results of the comparison methods. Visually, it can be observed that SDP shows good performance in terms of color and brightness.

IV. CONCLUSION

This article introduces an SDP method for FusHSISR, leveraging spectral distribution priors within a diffusion model framework. First, a novel SDM is proposed to depict spectral forward diffusion and reverse generation processes, enabling a tailored MLP-based denoising network for 1-D spectral data. Second, an unsupervised fusion model is designed to integrate SDM as a prior, enforcing target HR-HSI adherence to learned spectral distribution. To achieve this, the reverse spectral generation utilizes Markov state transitions as constraints, optimized via KL divergence-based loss. Finally, Adam optimization sequentially solves the corresponding subproblems in reverse generation order. Experiments on synthetic and real data validate SDP's effectiveness, with FID indicating generation quality's impact on fusion.

Despite the highly promising results achieved through SDP, there remains ample room for further refinement and augmentation in future endeavors. Potential avenues for improvement include leveraging spatial information in the design of SDM to bolster the robustness of spectrum generation, exploring optimal blends of gradient descent methods with accelerated generation techniques to expedite the model's solution speed,

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and extending the application of SDM into the realm of generating high-quality HSIs.

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REFERENCES

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] L. Loncan et al., "Hyperspectral pansharpening: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[3] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 2, pp. 29–56, Jun. 2017.

[4] R. Dian, S. Li, B. Sun, and A. Guo, "Recent advances and new guidelines on hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 69, pp. 40–51, May 2021.

[5] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.

[6] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.

[7] X. Meng, H. Shen, H. Li, L. Zhang, and R. Fu, "Review of the pansharpening methods for remote sensing images based on the idea of meta-analysis: Practical discussion and challenges," *Inf. Fusion*, vol. 46, pp. 102–113, Mar. 2019.

[8] G. Vivone et al., "A new benchmark based on recent advances in multispectral pansharpening: Revisiting pansharpening with classical and emerging pansharpening methods," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 53–81, Mar. 2021.

[9] T.-M. Tu, S.-C. Su, H.-C. Shyu, and P. S. Huang, "A new look at IHS-like image fusion methods," *Inf. Fusion*, vol. 2, no. 3, pp. 177–186, Sep. 2001.

[10] F. Nencini, A. Garzelli, S. Baronti, and L. Alparone, "Remote sensing image fusion using the curvelet transform," *Inf. Fusion*, vol. 8, no. 2, pp. 143–156, Apr. 2007.

[11] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[12] C. Lanaras, E. Baltsavias, and K. Schindler, "Hyperspectral super-resolution by coupled spectral unmixing," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 3586–3594.

[13] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[14] J. Liu, Z. Wu, L. Xiao, J. Sun, and H. Yan, "A truncated matrix decomposition for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 29, pp. 8028–8042, 2020.

[15] W. Dong et al., "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.

[16] M. A. Veganzones, M. Simões, G. Licciardi, N. Yokoya, J. M. Bioucas-Dias, and J. Chanussot, "Hyperspectral super-resolution of locally low rank images from complementary multisource data," *IEEE Trans. Image Process.*, vol. 25, no. 1, pp. 274–288, Jan. 2016.

[17] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[18] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Hyperspectral images super-resolution via learning high-order coupled tensor ring representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 11, pp. 4747–4760, Nov. 2020.

[19] Y. Chen, J. Zeng, W. He, X.-L. Zhao, and T.-Z. Huang, "Hyperspectral and multispectral image fusion using factor smoothed tensor ring decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515417.

[20] F. Ye, Z. Wu, X. Jia, J. Chanussot, Y. Xu, and Z. Wei, "Bayesian nonlocal patch tensor factorization for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 32, pp. 5877–5892, 2023.

[21] L. Zhang, W. Wei, C. Bai, Y. Gao, and Y. Zhang, "Exploiting clustering manifold structure for hyperspectral imagery super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 5969–5982, Dec. 2018.

[22] J. Xue, Y.-Q. Zhao, Y. Bu, W. Liao, J. C. Chan, and W. Philips, "Spatial–spectral structured sparse low-rank representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 3084–3097, 2021.

[23] R. Dian and S. Li, "Hyperspectral image super-resolution via subspace-based low tensor multi-rank regularization," *IEEE Trans. Image Process.*, vol. 28, no. 10, pp. 5135–5146, Oct. 2019.

[24] Q. Yuan, Y. Wei, X. Meng, H. Shen, and L. Zhang, "A multiscale and multidepth convolutional neural network for remote sensing imagery pan-sharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 978–989, Mar. 2018.

[25] Q. Li, Y. Yuan, and Q. Wang, "Multiscale factor joint learning for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5523110.

[26] Y. Zhang, C. Liu, M. Sun, and Y. Ou, "Pan-sharpening using an efficient bidirectional pyramid network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5549–5563, Aug. 2019.

[27] J. Jiang, H. Sun, X. Liu, and J. Ma, "Learning spatial–spectral prior for super-resolution of hyperspectral imagery," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 1082–1096, 2020.

[28] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial–spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5953–5965, Jul. 2021.

[29] X. Dong, X. Sun, X. Jia, Z. Xi, L. Gao, and B. Zhang, "Remote sensing image super-resolution using novel dense-sampling networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1618–1633, Feb. 2021.

[30] W. Xie, J. Lei, Y. Cui, Y. Li, and Q. Du, "Hyperspectral pansharpening with deep priors," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1529–1543, May 2020.

[31] X. Fu, W. Wang, Y. Huang, X. Ding, and J. Paisley, "Deep multiscale detail networks for multiband spectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 5, pp. 2090–2104, May 2021.

[32] Q. Li, M. Gong, Y. Yuan, and Q. Wang, "Symmetrical feature propagation network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536912.

[33] Q. Li, Y. Yuan, X. Jia, and Q. Wang, "Dual-stage approach toward hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 31, pp. 7252–7263, 2022.

[34] J. Li et al., "Hyperspectral image super-resolution by band attention through adversarial learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4304–4318, Jun. 2020.

[35] W. Dong, S. Hou, S. Xiao, J. Qu, Q. Du, and Y. Li, "Generative dual-adversarial network with spectral fidelity and spatial enhancement for hyperspectral pansharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 7303–7317, Dec. 2022.

[36] Q. Xie, M. Zhou, Q. Zhao, Z. Xu, and D. Meng, "MHF-Net: An interpretable deep network for multispectral and hyperspectral image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 3, pp. 1457–1473, Mar. 2022.

[37] W. Dong, J. Qu, S. Xiao, T. Zhang, Y. Li, and X. Jia, "Noise prior knowledge informed Bayesian inference network for hyperspectral super-resolution," *IEEE Trans. Image Process.*, vol. 32, pp. 3121–3135, 2023.

[38] W. G. C. Bandara and V. M. Patel, "HyperTransformer: A textural and spectral feature fusion transformer for pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1757–1767.

[39] K. Wang, X. Liao, J. Li, D. Meng, and Y. Wang, "Hyperspectral image super-resolution via knowledge-driven deep unrolling and transformer embedded convolutional recurrent neural network," *IEEE Trans. Image Process.*, vol. 32, pp. 4581–4594, 2023.

[40] S. Jia, Z. Min, and X. Fu, "Multiscale spatial–spectral transformer network for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 96, pp. 117–129, Aug. 2023.

[41] C. Li et al., "CasFormer: Cascaded transformers for fusion-aware computational hyperspectral imaging," *Inf. Fusion*, vol. 108, Aug. 2024, Art. no. 102408.

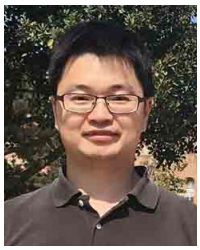
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[42] S. Li, S. Li, and L. Zhang, "Hyperspectral and panchromatic images fusion based on the dual conditional diffusion models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526315.
[43] C. Wu, D. Wang, Y. Bai, H. Mao, Y. Li, and Q. Shen, "HSR-diff: Hyperspectral image super-resolution via conditional diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 7060–7070.
[44] Q. Meng, W. Shi, S. Li, and L. Zhang, "PanDiff: A novel pansharpening method based on denoising diffusion probabilistic model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5611317.
[45] Z. Cao, S. Cao, L.-J. Deng, X. Wu, J. Hou, and G. Vivone, "Diffusion model with disentangled modulations for sharpening multispectral and hyperspectral images," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102158.
[46] B. Zhang et al., "Progress and challenges in intelligent remote sensing satellite systems," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1814–1822, 2022.
[47] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.
[48] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NIPS*, vol. 33. Vancouver, BC, Canada: Curran Associates, 2020, pp. 6840–6851.
[49] L. Yang et al., "Diffusion models: A comprehensive survey of methods and applications," *ACM Comput. Surveys*, vol. 56, no. 4, pp. 1–39, Apr. 2024.
[50] L. Zhang, J. Nie, W. Wei, Y. Li, and Y. Zhang, "Deep blind hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2388–2400, Jun. 2021.
[51] R. Dian, S. Li, and X. Kang, "Regularizing hyperspectral and multispectral image fusion by CNN denoiser," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1124–1135, Mar. 2021.
[52] X. Rui, X. Cao, L. Pang, Z. Zhu, Z. Yue, and D. Meng, "Unsupervised hyperspectral pansharpening via low-rank diffusion model," *Inf. Fusion*, vol. 107, Jul. 2024, Art. no. 102325.
[53] Y. Qu, H. Qi, and C. Kwan, "Unsupervised sparse Dirichlet-Net for hyperspectral image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2511–2520.
[54] Z. Wang, B. Chen, R. Lu, H. Zhang, H. Liu, and P. K. Varshney, "FusionNet: An unsupervised convolutional variational network for hyperspectral and multispectral image fusion," *IEEE Trans. Image Process.*, vol. 29, pp. 7565–7577, 2020.
[55] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509417.
[56] J. Liu, Z. Wu, L. Xiao, and X.-J. Wu, "Model inspired autoencoder for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522412.
[57] Y. Chen, X. Gui, J. Zeng, X.-L. Zhao, and W. He, "Combining low-rank and deep plug-and-play priors for snapshot compressive imaging," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jul. 21, 2024, doi: 10.1109/TNNLS.2023.3294262.
[58] Y. Chen, W. Lai, W. He, X.-L. Zhao, and J. Zeng, "Hyperspectral compressive snapshot reconstruction via coupled low-rank subspace representation and self-supervised deep network," *IEEE Trans. Image Process.*, vol. 33, pp. 926–941, 2024.
[59] Z. Kong, W. Ping, J. Huang, K. Zhao, and B. Catanzaro, "DiffWave: A versatile diffusion model for audio synthesis," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 1–12.
[60] C. Niu et al., "Permutation invariant graph generation via score-based generative modeling," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2020, pp. 4474–4484.
[61] J. Ho, T. Salimans, A. Gritsenko, W. Chan, M. Norouzi, and D. J. Fleet, "Video diffusion models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 8633–8646.
[62] A. Graikos, N. Malkin, N. Jojic, and D. Samaras, "Diffusion models as plug-and-play priors," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 14715–14728.
[63] H. Chung, J. Kim, M. T. Mccann, M. L. Klasky, and J. C. Ye, "Diffusion posterior sampling for general noisy inverse problems," in *Proc. Int. Conf. Learn. Represent.*, 2023, pp. 1–9. [Online]. Available: https://openreview.net/forum?id=OnD9zGAGT0k
[64] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst. (NIPS)*. Red Hook, NY, USA: Curran Associates, 2017, pp. 6629–6640.
[65] L. Alparone, B. Aiazzi, S. Baronti, A. Garzelli, F. Nencini, and M. Selva, "Multispectral and panchromatic data fusion assessment without reference," *Photogrammetric Eng. Remote Sens.*, vol. 74, no. 2, pp. 193–200, Feb. 2008.
[66] T. Ranchin and L. Wald, "Fusion of high spatial and spectral resolution images: The arsis concept and its implementation," *Photogramm. Eng. Remote Sens.*, vol. 66, no. 1, pp. 49–61, 2000.
[67] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Hyperspectral and multispectral image fusion via deep two-branches convolutional neural network," *Remote Sens.*, vol. 10, no. 5, p. 800, May 2018.



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